Katja Glass Consulting

Navigating Open Seas

Exploring Open Source Technology in Clinical Data Analysis (OSTCDA)

Disclaimer

These opinions are our own and do not necessarily reflect any connected companies position or strategy.

Agenda

- Introduction
- Manuscript
- Collaboration
- Initiatives
- Summary

Introduction

Introduction

- Open Source Technology Director, PHUSE
- How can we support OS in our industry
  - Uncertainty
  - Questions
  - Need for information
- Need a manuscript!

Introduction

- Open Source Technology in Clinical Data Analysis (OSTCDA)
  - PHUSE group

Michael Rimler
Open Source Technologies Director, PHUSE

Katja Glaß
Katja Glass Consulting
COSA Board Member

Mike Stackhouse
Chief Innovation Officer, Atorus
PHUSE DVOST Working Group co-Lead

Mike Smith
Lead of R Centre of Excellence, Pfizer

Introduction

Purpose

To characterize what we know with respect to ***embedding open-source solutions*** into all matters of clinical data handling, reporting, and submissions

Content

- Broad set of topics
- Comprehensive
  - Quoted content from industry experts
  - Resource lists and links
  - View of the future
- Citable / referenceable answers where possible

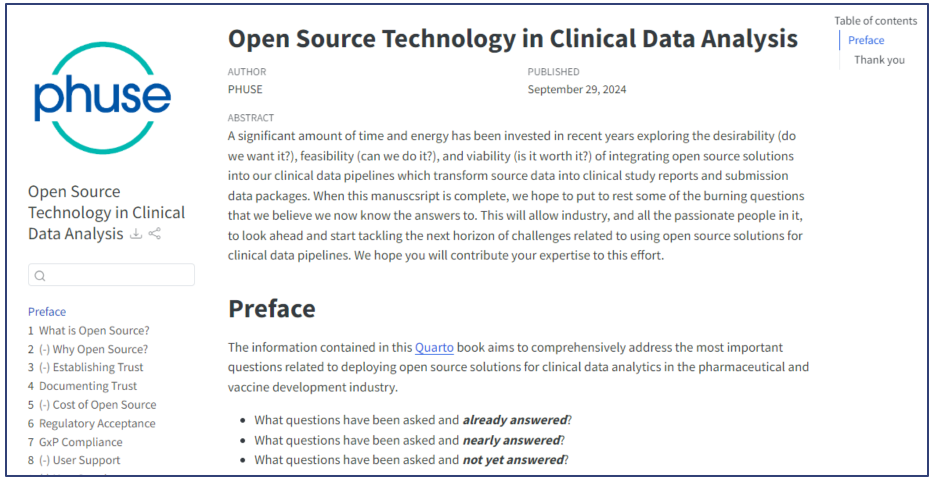
Structure

- Introduction
- What have we asked…
  - and answered?
  - and nearly answered?
  - and not yet answered?
- What have we not yet asked?
- Conclusions

Manuscript

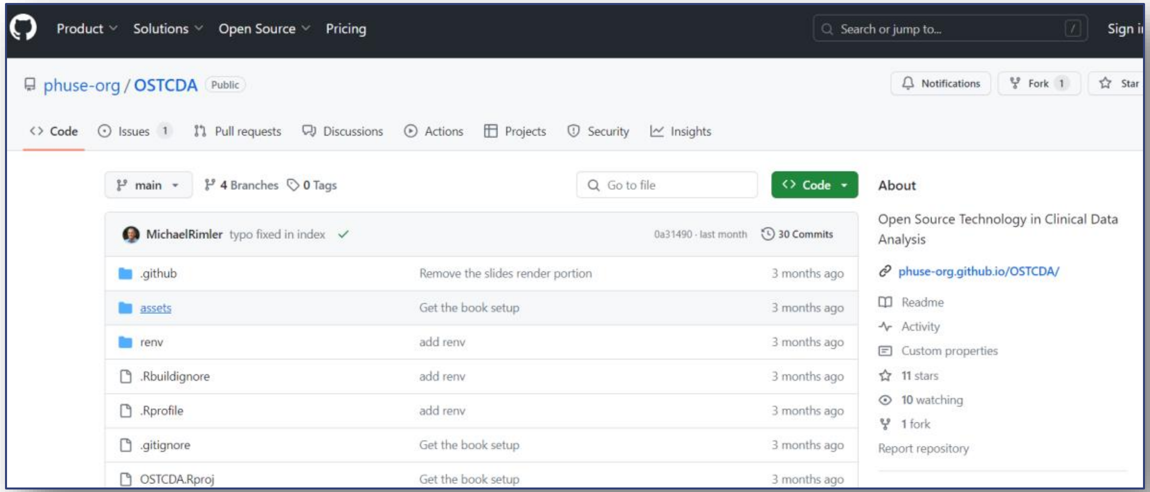
Manuscript

Manuscript:



https://phuse-org.github.io/OSTCDA/

Repository:



https://github.com/phuse-org/OSTCDA

[DISCUSSIONS]()

[ISSUES / FEEDBACK]()

Manuscript

- What is Open Source?
- What is the ‘why’ for using open source solutions in pharma clinical data analytics?
- How do you document your trust in an open source solution?
- Will the regulatory agencies accept data and analyses generated with solutions developed and available as open source?

Manuscript

1 What is Open Source?
2 (-) Why Open Source?
3 (-) Establishing Trust
4 Documenting Trust
5 (-) Cost of Open Source
6 Regulatory Acceptance
7 GxP Compliance
8 (-) User Support

9 (-) User Development
10 Numerical Matching
11 (-) OS in the Long Run
12 (-) Funding OS
13 Liability with OS
14 Legal Concerns
15 (-) OS Business Models
16 (-) What Else?

Manuscript

6 Regulatory Acceptance

6.1 Will the regulatory agencies accept data and analyses generated with solutions developed and available as open source?

- What do we know regarding data submissions to FDA?
- What do we know regarding data submissions to other regulatory agencies?
- Are there technical considerations for the creation of submission data packages?

6.2 (Perceived) Fear

There has been a long debate about whether regulatory agencies would accept submissions using R and Open Source tools. This is in spite of communications from agency staff for ***over 15 years*** refuting this and stating that the agency would not and could not endorse any specific software tool for sponsor

Table of contents

6.1 Will the regulatory agencies accept data and analyses generated with solutions developed and available as open source?
6.2 (Perceived) Fear
6.3 (Reducing) Uncertainty
6.4 (Avoiding) Doubt
6.5 Industry Experiences
6.6 Discussion
6.7 References

Manuscript

Regulartory Acceptance

- Authority: „vendor-independent“
- Industry experiences
  - Need more!
    Include / Tell us

Sponsor	Agency	Experiences
RConsortium Pilot Info	FDA	Submitted • Accepted
Novo Nordisk Presentation	FDA	Submitted ...
	EMA	Submitted content using R
	NMPA	Same as PMDA
	PMDA	Submitted ...
	Health Canada	Submitted content using R
Roche Presentation	FDA	Submitted content using R ...
	EMA	Submitted content using R ...
	NMPA	Submitted content using R ...
Merck	FDA	Submitted ...
Astellas Pharma	FDA	(2021) for using R markdown to submit programs Example

Manuscript

7 GxP Compliance

7.1 How do you establish accuracy, reproducibility and traceability?

FDA definition of validation and GxP compliance means establishing accuracy, reproducibility, and traceability. When working with open source solutions to process and analyze clinical trial data:

- How do we establish reproducibility of the outputs?
- How do we establish traceability of the input through to the output?

7.2 Are we validating R packages or the R environment?

When discussing R validation, most of the discussion centres around validation of R functions and packages individually. Do they give the "right" or "expected" answer within a defined tolerability / accuracy? Can we trust the developers of the packages that they have adequately tested and documented

Table of contents

7.1 How do you establish accuracy, reproducibility and traceability?
7.2 Are we validating R packages or the R environment?
7.3 R Validation Hub and \{riskmetric\}
7.4 R package management - the problem
7.5 R package management - possible solutions
7.6 Traceability - Where is the R log?
7.7 The ideal solution
7.8 Discussion

Manuscript

- R Validation Hub & {riskmetrix}
  - Testing
  - Documentation
- Package / environment management
- Traceability tools

Manuscript

13 Liability with OS

13.1 Are contributors to open source exposing themselves to any liability of their solutions?

- What are possible sources of liabilities?
- Are there mitigating actions which can limit or eliminate liabilities?

13.2 Understanding Liability and Warranty in Software Development

Liability and warranty are two crucial legal concepts in the realm of software development. **Liability** refers to the legal obligation or responsibility that arises

Table of contents

13.1 Are contributors to open source exposing themselves to any liability of their solutions?

13.2 Understanding Liability and Warranty in Software Development

13.3 Liability and Warranty according Open Source Licenses

13.4 Discussion

Manuscript

- Uncertainty

- Risk is on sponsor
  - Not from the person who has created
  - Neither vendor, CRO, nor open-source provider
  - Similar to copy from a colleague

- Licenses clarify this
  - No warranty, no liability
  - All do their best

Manuscript

14 Legal Concerns

14.1 Are there any legal concerns from open source development?

- What do individuals need to know?
- What do organizations need to know?
- How does this differ if the solution is an individual or a collaborative effort?

14.2 Understanding Legal Concerns

In our world today, where claims are frequently made, it's clear that many of us have concerns about the potential legal issues tied to open source development. Whether we're contributing as individuals or as part of an organization, we all have a role to play. It's only natural and indeed obvious that we might feel unsure or anxious about possible missteps. However, by deepening our understanding of these legal

Table of contents

14.1 Are there any legal concerns from open source development?
14.2 Understanding Legal Concerns
14.3 Legal Practicalities
14.4 Additional Insights
14.5 Discussion

Manuscript

- Understanding license terms
  - No license -> copyright

Tips:

- Check for license
- Prefer permissible licenses (MIT, Apache)
- Rewrite instead of copy
- Maintain a list of open source software + licenses

Collaboration

Collaboration

- Interviews

- Call for participation

- Discussion meetings

Collaboration

- Open Forum Discussions

26 April

17 May

(CSS) 4 June

14 June

12 July

13 Sept.

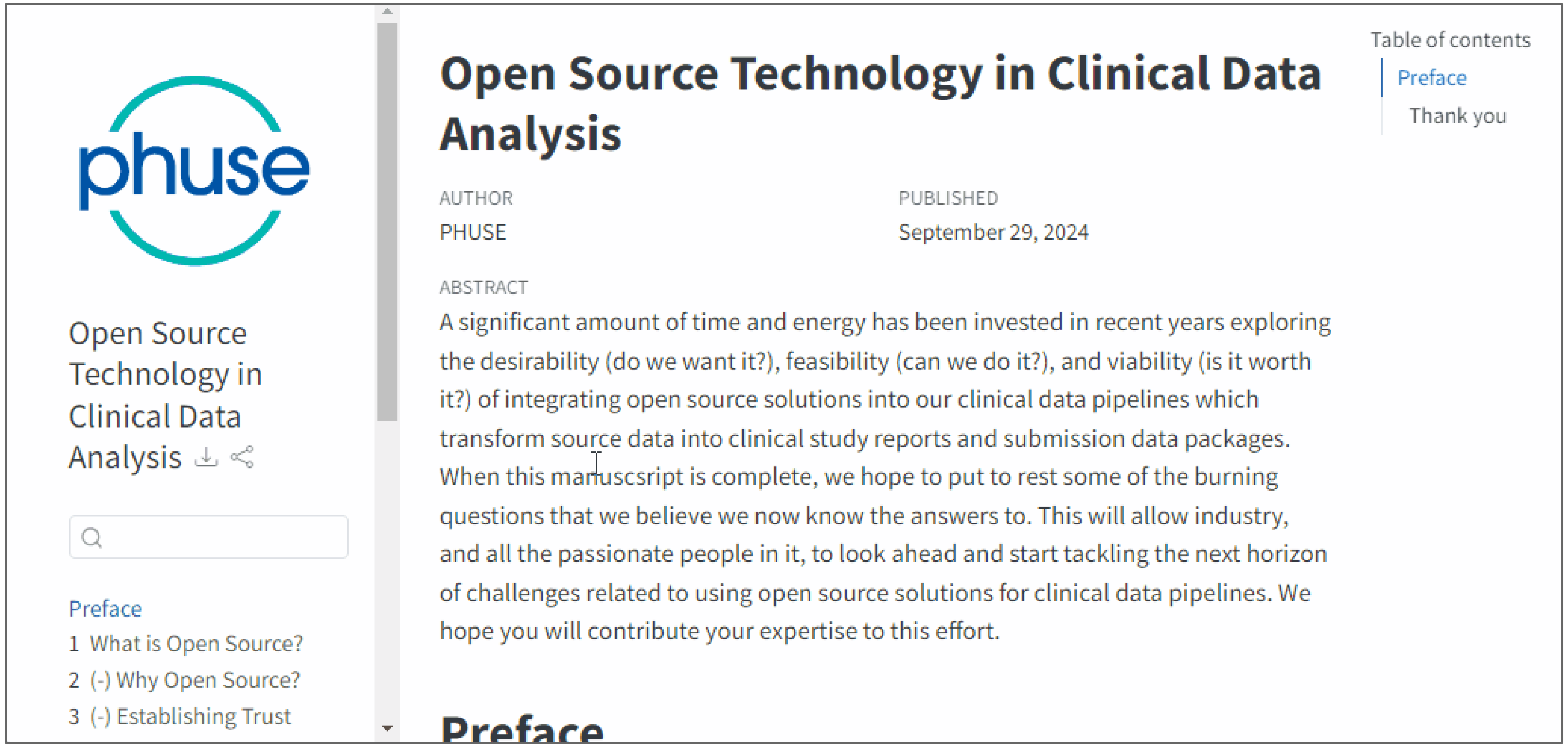
11 Oct.

13 Nov.

PHUSE EU Connect

Collaboration

- Vote / comment on discussions & feedback (Issues)



phuse

Open Source Technology in Clinical Data Analysis

Preface
1 What is Open Source?
2 (-) Why Open Source?
3 (-) Establishing Trust

Open Source Technology in Clinical Data Analysis

AUTHOR
PHUSE

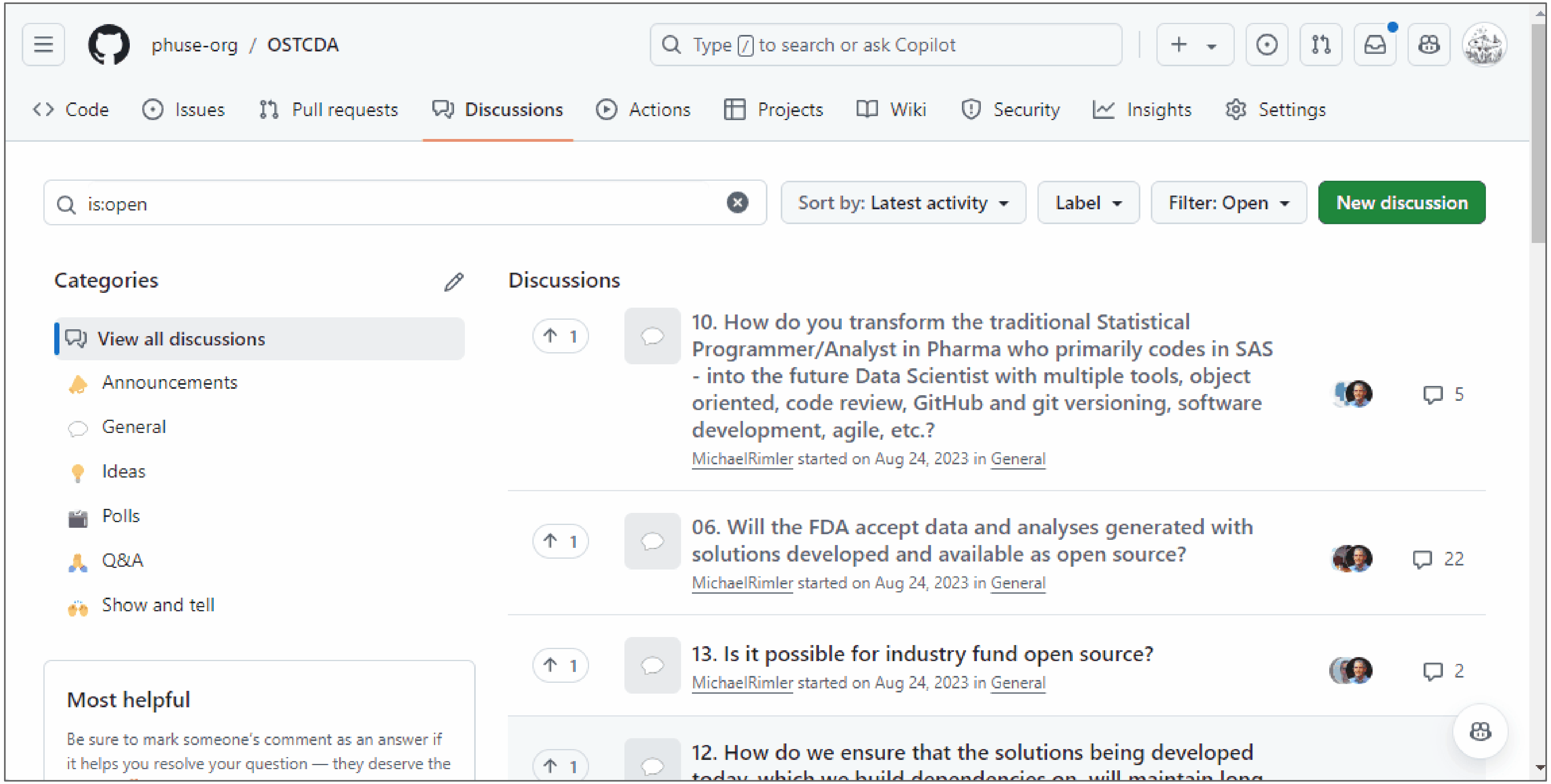
PUBLISHED
September 29, 2024

ABSTRACT
A significant amount of time and energy has been invested in recent years exploring the desirability (do we want it?), feasibility (can we do it?), and viability (is it worth it?) of integrating open source solutions into our clinical data pipelines which transform source data into clinical study reports and submission data packages. When this manuscsript is complete, we hope to put to rest some of the burning questions that we believe we now know the answers to. This will allow industry, and all the passionate people in it, to look ahead and start tackling the next horizon of challenges related to using open source solutions for clinical data pipelines. We hope you will contribute your expertise to this effort.

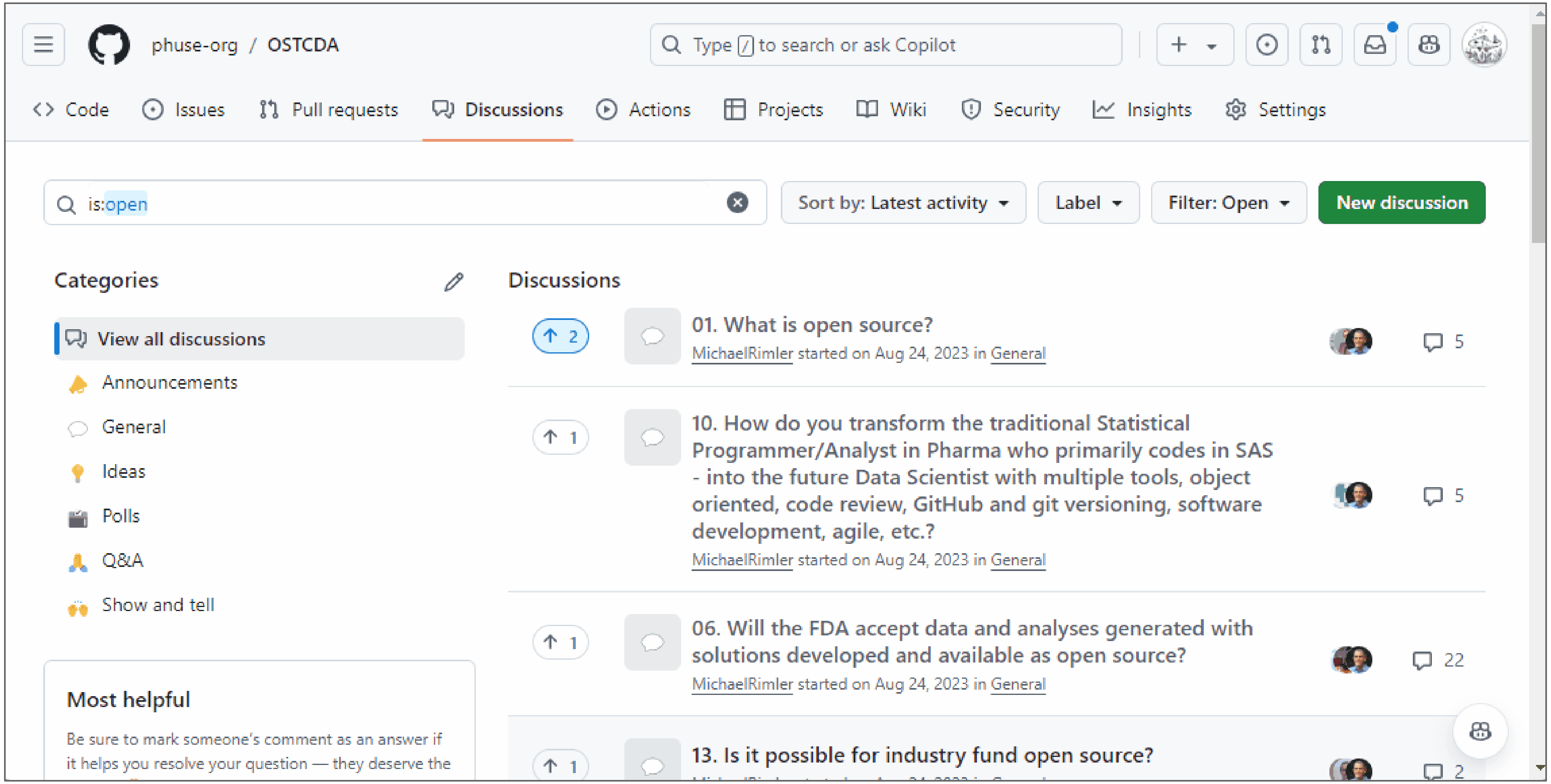
Preface

Table of contents
Preface
Thank you

Collaboration



Collaboration



Collaboration

Interactive Session, Wednesday 13th, 11:00-12:30

- Present OSTCDA working group
- Discuss on topics

Collaboration

- Open source at work
- Using R at work
- Collaborate in open source project
- Develop/maintain open source project

Initiatives

Initiatives

- PHUSE OSTCDA
- CDISC Open Source Alliance (COSA)
- Pharmaverse

Initiatives

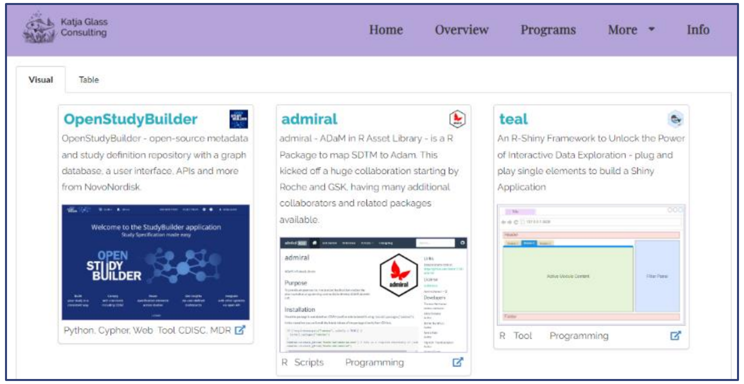
- R Consortium
- R Adoption Series
- R Pilot Submission

https://rconsortium.github.io/submissions-wg/

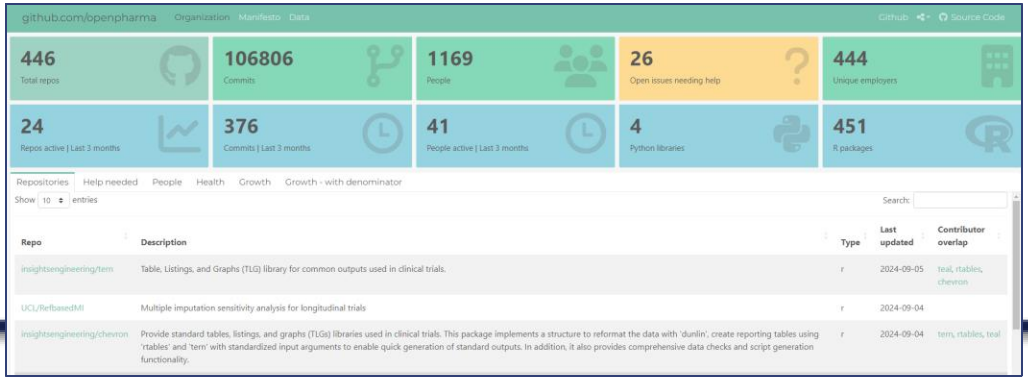
- R Validation Hub

Initiatives

- Open Source Portal (www.glacon.eu/portal)

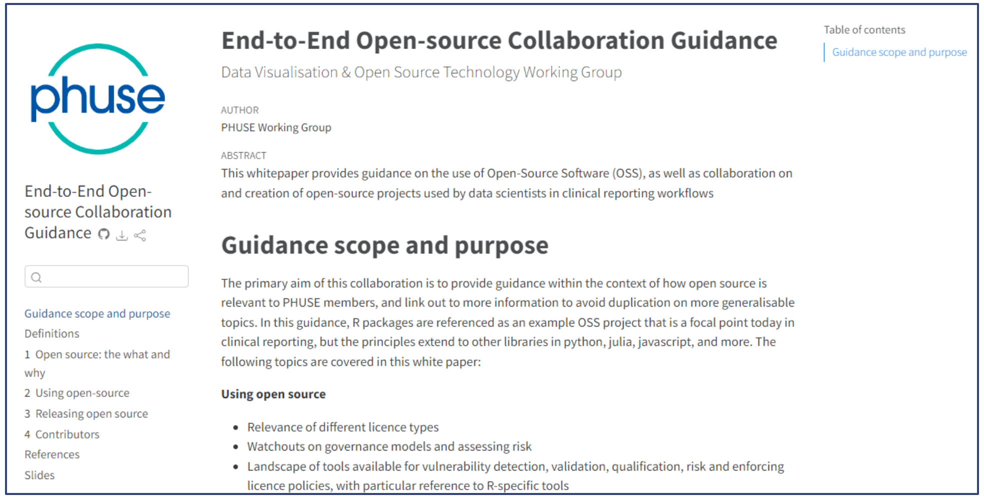


- Open Pharma (https://openpharma.github.io/)



Initiatives

- End-2-End Guide
  (https://phuse-org.github.io/E2E-OS-Guidance/)



phuse

End-to-End Open-source Collaboration Guidance

Guidance scope and purpose
Definitions
1 Open source: the what and why
2 Using open-source
3 Releasing open source
4 Contributors
References
Slides

End-to-End Open-source Collaboration Guidance

Data Visualisation & Open Source Technology Working Group

AUTHOR
PHUSE Working Group

ABSTRACT
This whitepaper provides guidance on the use of Open-Source Software (OSS), as well as collaboration on and creation of open-source projects used by data scientists in clinical reporting workflows

Guidance scope and purpose

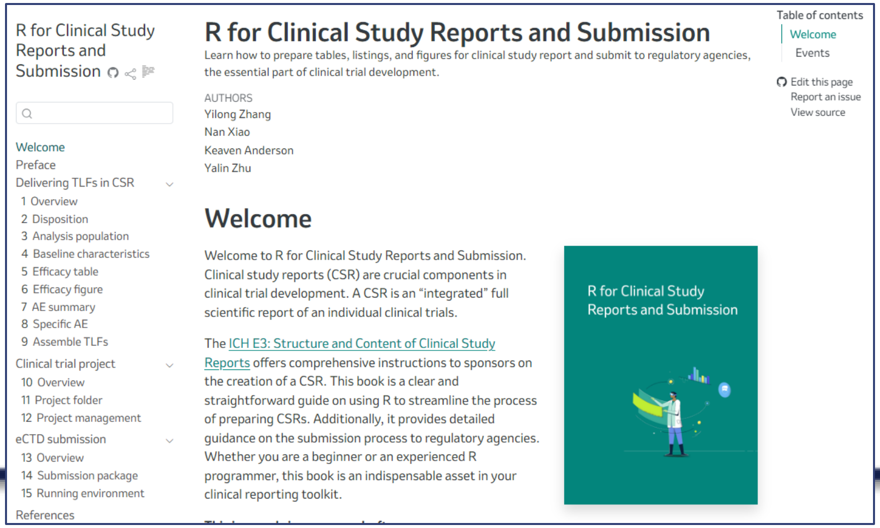
The primary aim of this collaboration is to provide guidance within the context of how open source is relevant to PHUSE members, and link out to more information to avoid duplication on more generalisable topics. In this guidance, R packages are referenced as an example OSS project that is a focal point today in clinical reporting, but the principles extend to other libraries in python, julia, javascript, and more. The following topics are covered in this white paper:

Using open source

- Relevance of different licence types
- Watchouts on governance models and assessing risk
- Landscape of tools available for vulnerability detection, validation, qualification, risk and enforcing licence policies, with particular reference to R-specific tools

Table of contents
Guidance scope and purpose

- R for Clinical Study Reports and Submission
  (https://r4csr.org/)



R for Clinical Study Reports and Submission

Welcome
Preface
Delivering TLFs in CSR
1 Overview
2 Disposition
3 Analysis population
4 Baseline characteristics
5 Efficacy table
6 Efficacy figure
7 AE summary
8 Specific AE
9 Assemble TLFs
Clinical trial project
10 Overview
11 Project folder
12 Project management
eCTD submission
13 Overview
14 Submission package
15 Running environment
References

R for Clinical Study Reports and Submission

Learn how to prepare tables, listings, and figures for clinical study report and submit to regulatory agencies, the essential part of clinical trial development.

AUTHORS
Yilong Zhang
Nan Xiao
Keaven Anderson
Yalin Zhu

Welcome

Welcome to R for Clinical Study Reports and Submission. Clinical study reports (CSR) are crucial components in clinical trial development. A CSR is an "integrated" full scientific report of an individual clinical trials.

The ICH E3: Structure and Content of Clinical Study Reports offers comprehensive instructions to sponsors on the creation of a CSR. This book is a clear and straightforward guide on using R to streamline the process of preparing CSRs. Additionally, it provides detailed guidance on the submission process to regulatory agencies. Whether you are a beginner or an experienced R programmer, this book is an indispensable asset in your clinical reporting toolkit.

Table of contents
Welcome
Events
Edit this page
Report an issue
View source

Initiatives

- Concrete projects

admiral

Dataset-JSON

RAMNOG

oak

OPEN STUDY BUILDER

...

Summary

Summary

- Contribute to the Discussions in the project repository
- 👍 Upvote answers you like
- Add questions we've missed
- Review the draft content and submit Pull Request to update content

Summary

- Industry expertise grows
  - More open-source experiences
  - Share lessons learned
  - Build knowledge base

- Enable opportunities through collaboration

- Add your comments & inputs

Questions?

Katja Glass Consulting

info@glacon.eu

www.glacon.eu

www.glacon.eu/portal